Lecture 30

Monday, 12 October 2026 8:24 AM

Temp: $\mathbb{R}^3 \longrightarrow \mathbb{R}$

$(0, 0, 1) \longrightarrow 20^\circ$

$(1, 2, 0) \longrightarrow 30^\circ$

Scalar field $f: \mathbb{R}^n \longrightarrow \mathbb{R}$

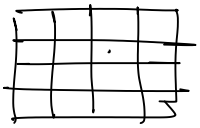
Image: $\mathbb{R}^2 \longrightarrow \mathbb{R}$

(x, y)

Vector field: $f: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ $(m > 1)$

Color image: $\mathbb{R}^2 \longrightarrow \mathbb{R}^3$

$(x, y) \longrightarrow (R, G, B)$



Q: How to find change in the scalar/vector field

Limit of a Scalar field:

$\lim f(\bar{x}) = L$

$\bar{x} \longrightarrow \bar{a}$

$f(\bar{x}) \longrightarrow L$ (finite) if $\bar{x} \longrightarrow \bar{a}$

ie $dist(\bar{x}, \bar{a}) \longrightarrow 0$

$\bar{x} = (x_1, x_2, \ldots x_n)$

$\bar{a} = (a_1, a_2, \ldots a_n)$

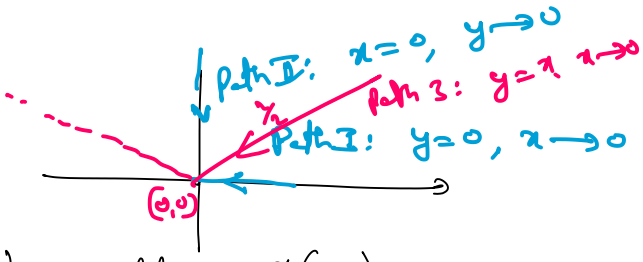
$dist(\bar{x}, \bar{a}) = \| \bar{x} - \bar{a} \|$

$= \sqrt{(x_1 - a_1)^2 + (x_2 - a_2)^2 + \ldots + (x_n - a_n)^2}$

Eg(1): $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = ?$

$f(x,y) = \frac{xy}{x^2+y^2}$

$f: \mathbb{R}^2 \longrightarrow \mathbb{R}$



Along Path I:

$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} \overset{\text{Path 1}}{=} \lim_{x\to 0} \frac{x(0)}{x^2+(0)^2}$

$= \lim_{x\to 0} \frac{0}{x^2} = 0$

Along Path II:

$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} \overset{\text{Path 2}}{=} \lim_{y\to 0} \frac{0(y)}{0^2+y^2} = 0$

As $\lim_{(x,y)\to(0,0)} f(x,y)$ is different in different directions, the limit doesn't exist.

Eg(2): $\lim_{(x,y)\to(0,0)} \frac{x^3 y}{x^6+y^2}$

Path 1: $y = x^3$, $x \to 0$

$\text{limit} = \frac{1}{2}$

Path 2: $y = 2x^3$, $x \to 0$

$\text{limit} = \frac{2}{5}$

Limit doesn't exist

∴ Limit

Eg3: $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$ (Feeling)

Means $\frac{xy}{\sqrt{x^2+y^2}} \to 0$ as $(x,y) \to (0,0)$

ie $\frac{xy}{\sqrt{x^2+y^2}} \to 0$ as $\|(x,y)-(0,0)\| \to 0$

ie as $\sqrt{x^2+y^2} \to 0$

Ex: $\lim_{(x,y)\to(0,0)} \frac{x-y}{x+y} = ?$